E-Laundry System

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Submission Form for Final-Year

PROJECT REPORT

Version	V X.0	Number Of Members	3

TITLE	E-Laundry System
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MEMBERS' SIGNATURES

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Supervisor's Signature

APPROVAL CERTIFICATE

This project, entitled as " E-Laundry System " has been approved for the award of

Bachelors of Software Engineering

Committee Signatures:

Supervisor: Ch Anwar Ul Hassan

Project Coordinator: Ch Anwar Ul Hassan

Head of Department: Dr. Imran Ahsan

DECLARATION

We, hereby, declare that "no portion of the work referred to in this project has been submitted in support of an application for another degree or qualification of this or any other university/institute or other institution of learning." It is further declared that this undergraduate project, neither as a whole nor as a part thereof, has been copied out from any sources, wherever references have been provided.

MEMBERS' SIGNATURES

Acknowledgements

We would like to express our heartfelt gratitude to all those who supported and guided us throughout the successful completion of our final year project,"E-Laundry System."

First and foremost, we are deeply thankful to our project supervisor, Ch,Anwar Ul Hassan, whose continuous guidance, insightful feedback, and constant motivation helped us stay on track and improve at every stage of the project. Their encouragement and expert advice were critical in shaping the outcome of this work.

We are also grateful to the faculty members of the CT Department, Air University, for equipping us with the technical knowledge and problem-solving skills that proved essential in this project.
Special thanks to our fellow classmates and friends who provided assistance, shared ideas, and offered feedback during development and testing phases.

This project is the result of the collaborative efforts of everyone mentioned above, and we are truly thankful for their contributions.

Executive Summary

Our project, *CleanEase: Online Laundry Management System*, aims to revolutionize the traditional laundry service industry by offering a modern, efficient, and user-friendly digital solution. This system enables users to schedule laundry pickups and deliveries, track order status in real-time, make secure online payments, and manage preferences—all through a web and mobile application.

Targeting urban professionals, students, and busy households, CleanEase addresses key pain points like time constraints, inconsistent service quality, and lack of transparency. For laundry service providers, it offers an easy-to-use backend to manage orders, customer data, inventory, and delivery logistics, leading to higher operational efficiency.

The platform supports a scalable business model with potential for partnerships with local laundromats and dry cleaners. With the growing demand for on-demand services and the shift towards digital convenience, CleanEase is well-positioned to capture a significant share of the laundry service market and expand into new regions.

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Chapter 1

Introduction

This project, titled “E-Laundry System,” aims to provide a digital platform for managing laundry services efficiently. The goal is to save users time by allowing them to schedule laundry pickups, choose services like washing or dry cleaning, and have their clothes delivered to their doorstep.

The system is designed for local laundry businesses and individual users who seek convenience in managing their laundry needs. It will help laundry service providers automate their operations and reach more customers online.

1.1. Project Introduction

In today’s fast-paced world, individuals often struggle to find time for basic household chores such as laundry. To address this, the E-Laundry System aims to provide a digital platform where users can schedule laundry pickups, track progress, make payments, and get their clothes delivered back—clean and fresh—without ever leaving their homes. The system is designed to streamline the laundry process for both customers and service providers, offering convenience, efficiency, and reliability.
The system allows users to schedule laundry pickups, choose services such as washing, ironing, and dry cleaning, and have their clothes delivered back to their doorstep. Through this platform, customers can track the status of their orders, make secure online payments, and communicate with the laundry service easily. The idea behind this project is to simplify daily life for users by digitizing the traditional laundry process. It also helps local laundry businesses to manage their operations more effectively by automating order tracking, customer management, and service delivery.

1.2. Existing Examples / Solutions

Several digital platforms already offer online laundry services, such as Laundryheap, DhobiLite, and TumbleDry. These systems typically provide features like online order placement, pickup scheduling, and delivery tracking. However, many of these platforms are limited to specific cities or lack affordability and customization options for small or local businesses. Our project addresses these gaps by offering a flexible, location-independent solution tailored to the needs of both customers and small-scale service providers.

1.3. Business Scope

This system has strong commercial potential, especially in urban and semi-urban areas where people are time-constrained. It can be deployed as a subscription-based or commission-based service model, allowing laundry shops to digitize and scale their operations. Potential customers include:

- Local laundromats
- Apartment complexes
- Busy professionals
- Students and families

Minimal additional resources like a delivery partner and basic server infrastructure can help launch the system commercially.

1.4. Useful Tools and Technologies

Below are the tools and technologies chosen for this project, along with brief justifications:

Layer	Technology Used
Frontend	HTML, CSS, JavaScript, React
Backend	Node.js / PHP / Django
Database	MySQL
Server/Hosting	Firebase / AWS / Local Server
Version Control	Git & GitHub
Design & Prototyping	Figma / Adobe XD

1.5. Project Work Break Down

The project is divided into tasks like UI design, development, database setup, and testing. Work is assigned based on each member's strengths, and more time is given to modules that need extra focus. This helps in managing the project efficiently.

Task	Duration	Team Member(s)	Remarks
Requirement Gathering & Analysis	1 Week	All	Collective effort to understand the scope and gather initial requirements.
UI/UX Design (Wireframes, Layout)	1 Week	Maham and Fatima	Strength in design. Tools like Figma used for better planning.
Frontend Development (Website)	2 Weeks	Pakeeza And Maham	Shared task. Pakeeza is strong in HTML/CSS, Maham supports JavaScript/React part.
Documentation & Report Writing	1 Week (parallel)	Fatima	Done alongside development; responsibilities split evenly.

1.6. Project Time Line

Phase	Duration	Timeline
Requirement Analysis	1 week	May 1 – May 7
Design Phase	1 week	May 8 – May 14
Frontend Development	2 weeks	May 15 – May 28
Backend Development	2 weeks	May 29 – June 11
Testing & Debugging	1 week	June 12 – June 18
Final Deployment & Report	1 week	June 19 – June 25

Chapter 2

Requirement Specification and Analysis

This document outlines the software requirements for the *Online Laundry System*. It is written in plain language to ensure clarity for both technical teams and non-technical stakeholders. The aim is to describe the functional and non-functional aspects of the system in a structured and understandable manner.

Requirement Specification:

Requirement specification is a crucial phase in the software development process that involves continuous communication with users and stakeholders to understand and define what the system must do. It helps in identifying user expectations, resolving conflicts or ambiguities in requirements, and documenting all aspects of the software from beginning to end.

For the E-Laundry System, the requirements describe the expected behavior of the application, including the features it must support and the attributes it must have. These requirements serve as a foundation for both the design and implementation phases.

2.1. Functional Requirements

The Functional Requirements Specification outlines the main operations and activities that the Online Laundry System must perform. It is written in simple language so that both technical and non-technical readers can clearly understand how the system works.

S. No.	Functional Requirement	Type	Status
1	User registration and login functionality	User Interaction	Implemented
2	Placing laundry service orders	Core Functionality	In Progress
3	Pickup and delivery scheduling	Workflow	Pending
4	Order tracking and status updates	Core Functionality	In Progress
5	Online payment integration	Financial	Pending
6	Admin dashboard for order/user management	Administrative	In Progress
7	Email/SMS notifications for order status	Communication	Pending
8	Generating order history and receipts	Reporting	Implemented

Table 2.1: Functional Requirements

2.2. Non-Functional Requirements

S. No.	Non-Functional Requirement	Category
1	The system should load all pages within 2–3 seconds on a standard internet connection	Performance
2	The application should support up to 100 concurrent users without crashing	Scalability
3	The platform should be accessible 24/7 with 99.9% uptime	Availability
4	User passwords and payment information must be securely stored using encryption techniques	Security
5	The system should be easily usable by new users without training	Usability
6	Errors and system failures must be logged and recoverable without data loss	Reliability
7	Admin should be able to update pricing or services without modifying source code	Maintainability

Table 2.2: Functional and Non-Functional Requirement

Chapter 3

3.1. Use Case

A use case defines a set of use-case instances, where each instance is a sequence of actions a system performs that yields an observable result of value to a particular actor. The functionality of a system is defined by different use cases, each of which represents a specific goal (to obtain the observable result of value) for a particular actor.

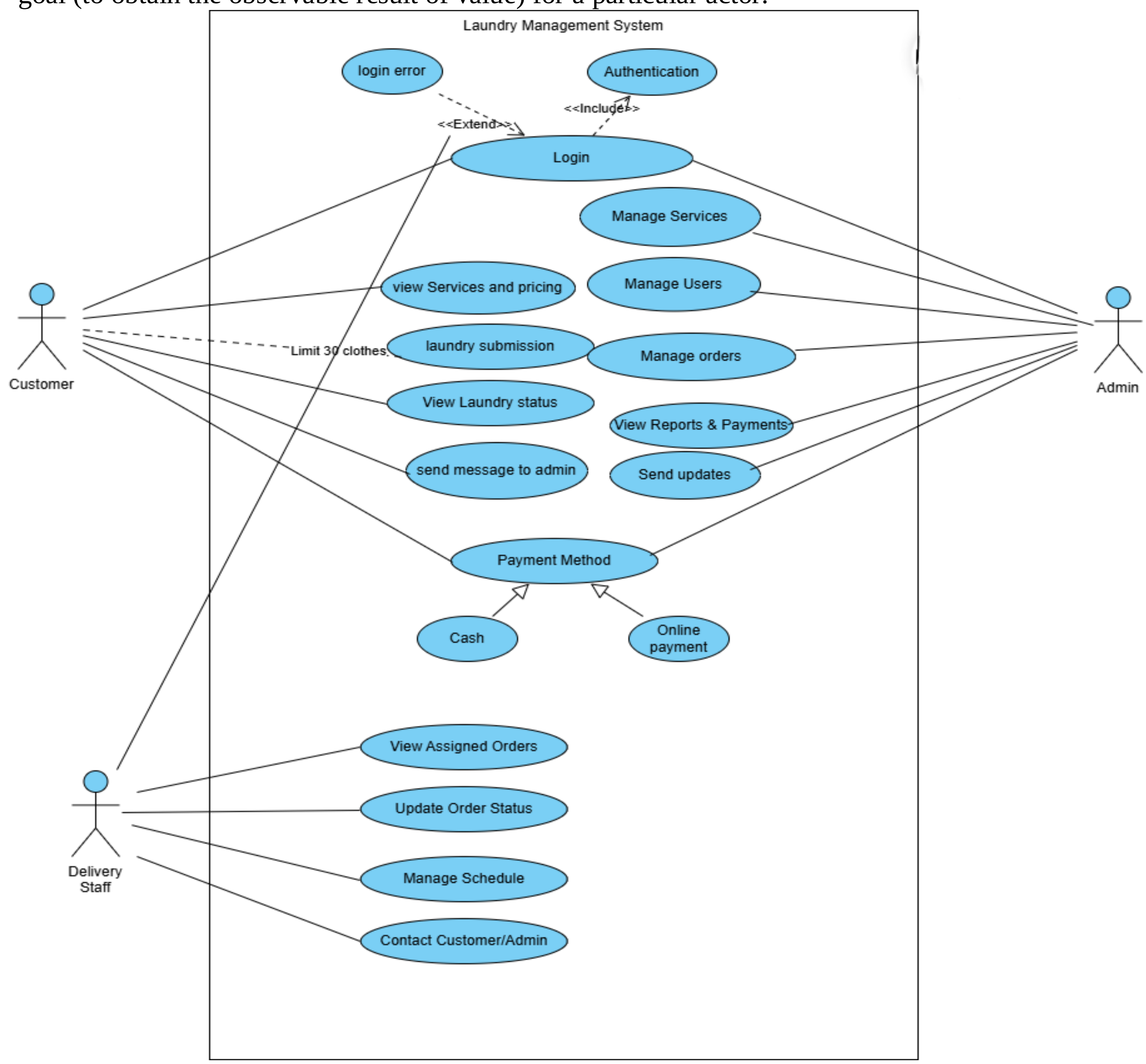

Figure 3.1. Use case Diagram

3.2. Fully Dressed Use Case

Field	Details
Use Case	Complete Laundry Order
Author	Fatima
Date	19-05-2025
Purpose	To place, manage, and complete a laundry order from request to delivery.
Overview	Customer logs in, places an order, admin processes it, delivery staff picks and delivers clothes.
Cross References	Login, Manage Orders, Assign Orders, Payment, Track Status
Actors	Customer, Admin, Delivery Staff
Pre-conditions	- Customer must be logged in- Services must be available- Delivery staff must be assigned
Post-conditions	- Laundry delivered- Payment made- Order marked as Completed
Normal Flow of Events	1. Customer logs in 2. Views services 3. Submits laundry order 4. Admin assigns delivery 5. Staff picks up laundry 6. Laundry cleaned 7. Delivery done 8. Customer pays 9. Admin verifies and completes order
Alternative Flow of Events	- Clothes exceed limit → System warns- Staff unavailable → Admin reschedules- Payment error → Retry needed
Exceptional Flow of Events	- Login failure- Delivery failed due to customer absence- Payment system downtime
Implementation Issues	- Delay in notifications- Integration with payment gateway- Mobile access for staff

3.3. ER Diagram

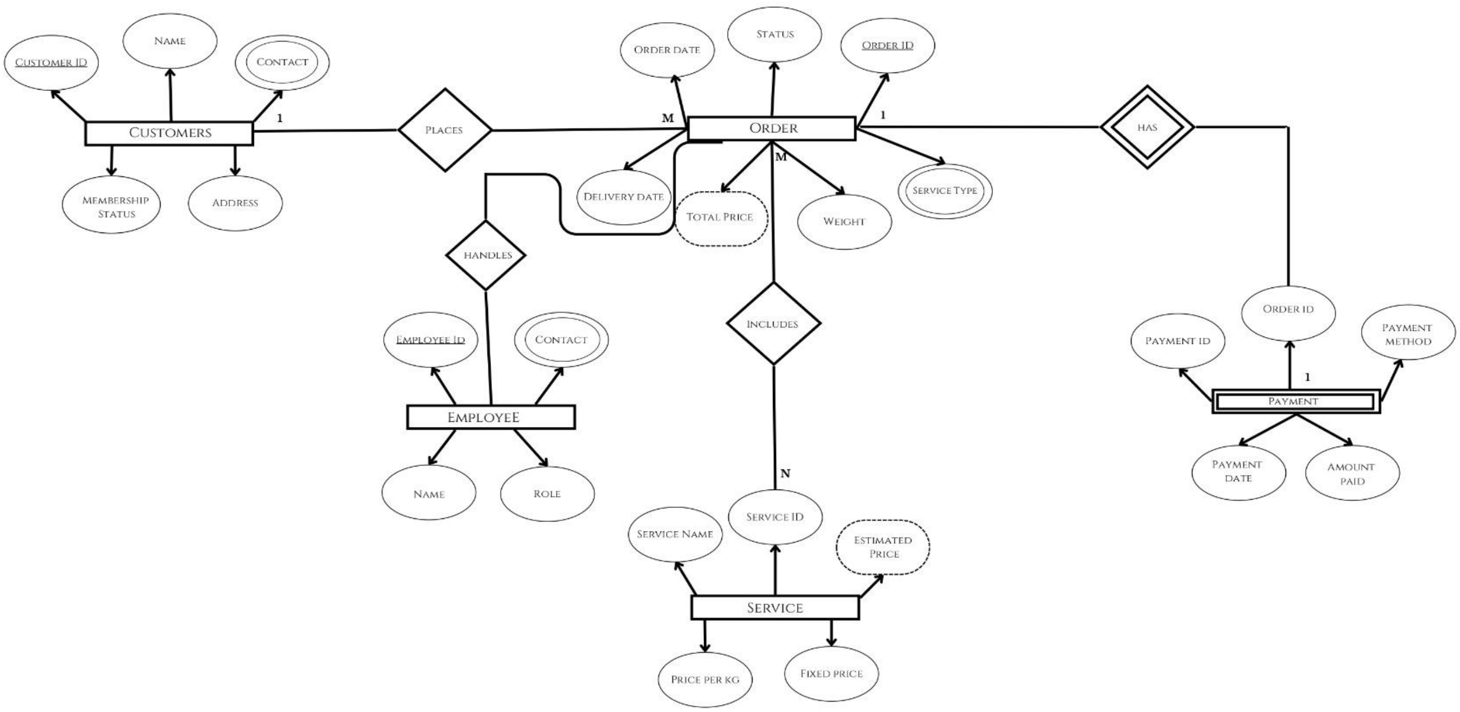

Figure 3.2. ER Diagram

3.4. Flow Chart

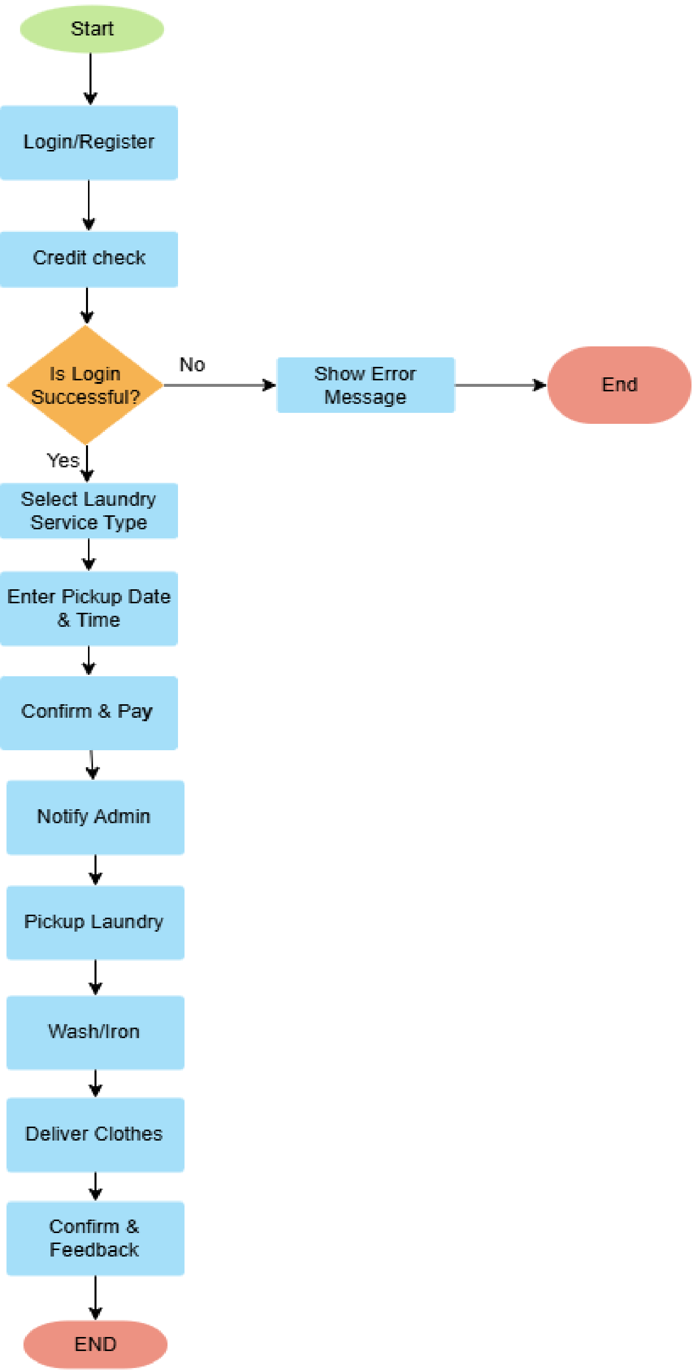

Figure 3.3.Flow Chart

3.5. Domain Model

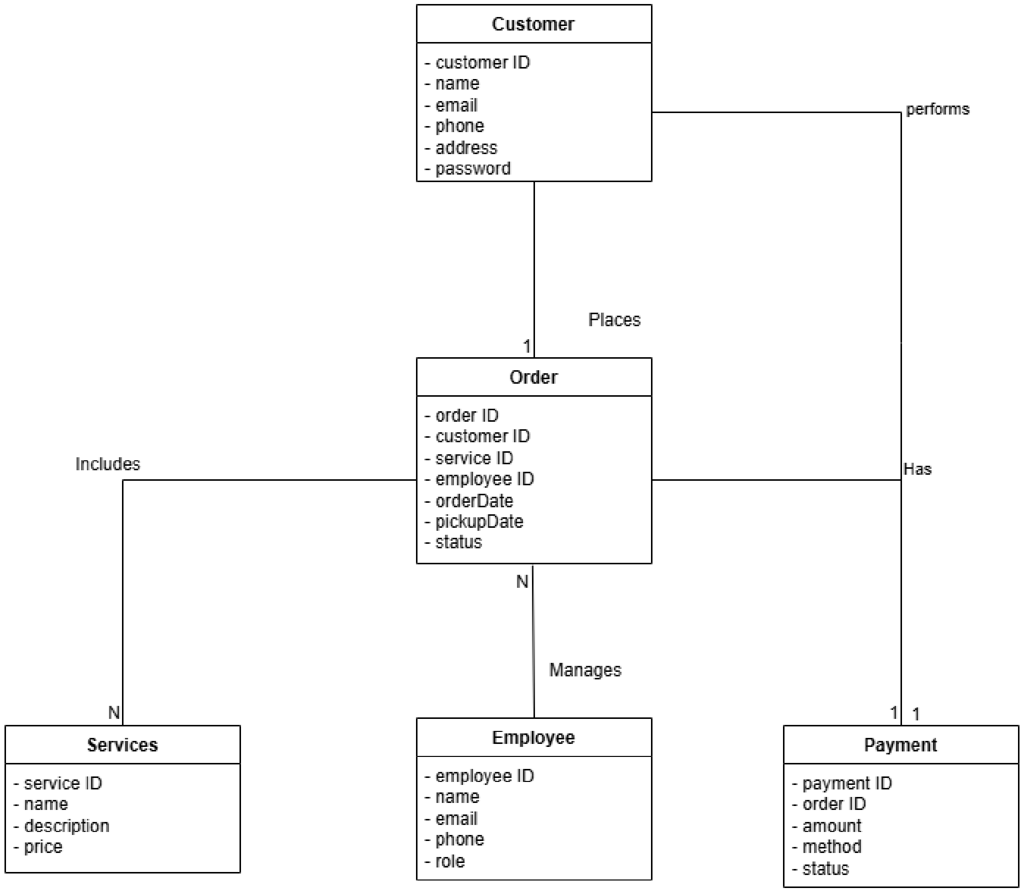

Figure 3.4. Domain Model

3.6. Class Diagram

Class Diagram as shown in Fig. 3.5 provides an overview of the target system by describing the objects and classes inside the system and the relationships between them. It provides a wide variety of usages; from modeling the domain-specific data structure to detailed design of the target system.

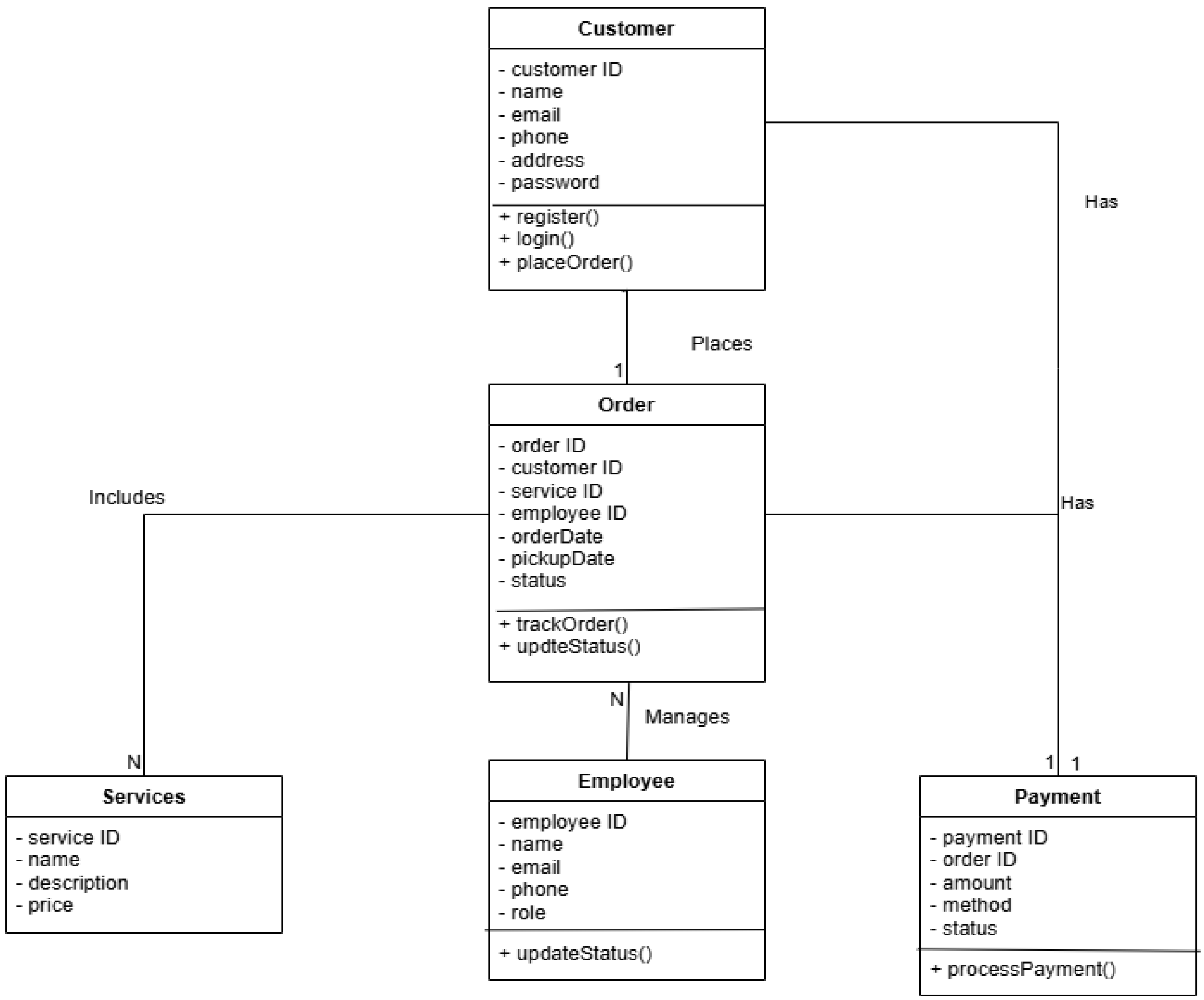

Figure 3.5.Class Diagram

3.7. System Sequence Diagram

1-Place Laundry Order

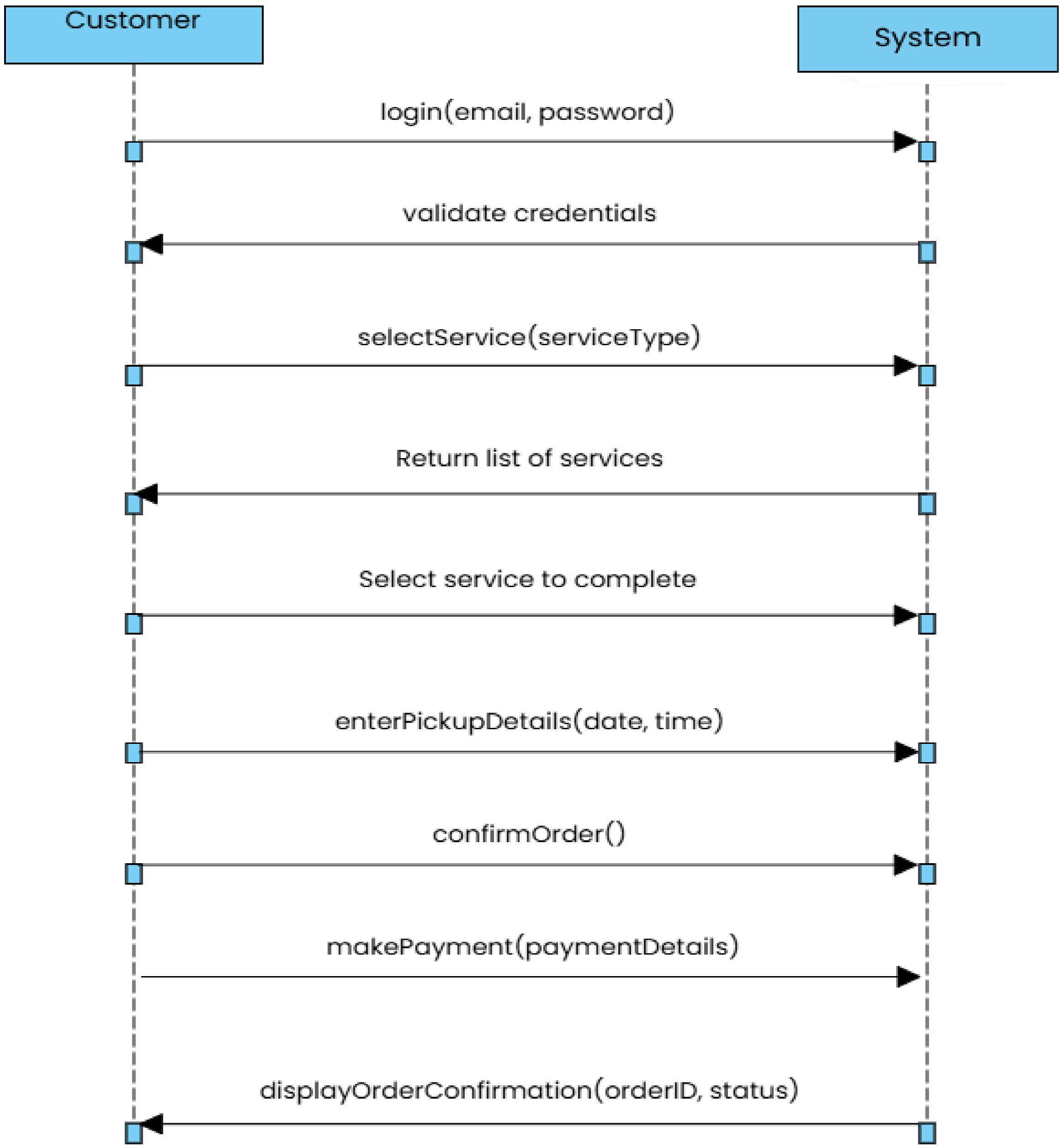

Figure 3.6.SSD for placing order

2-Track Order

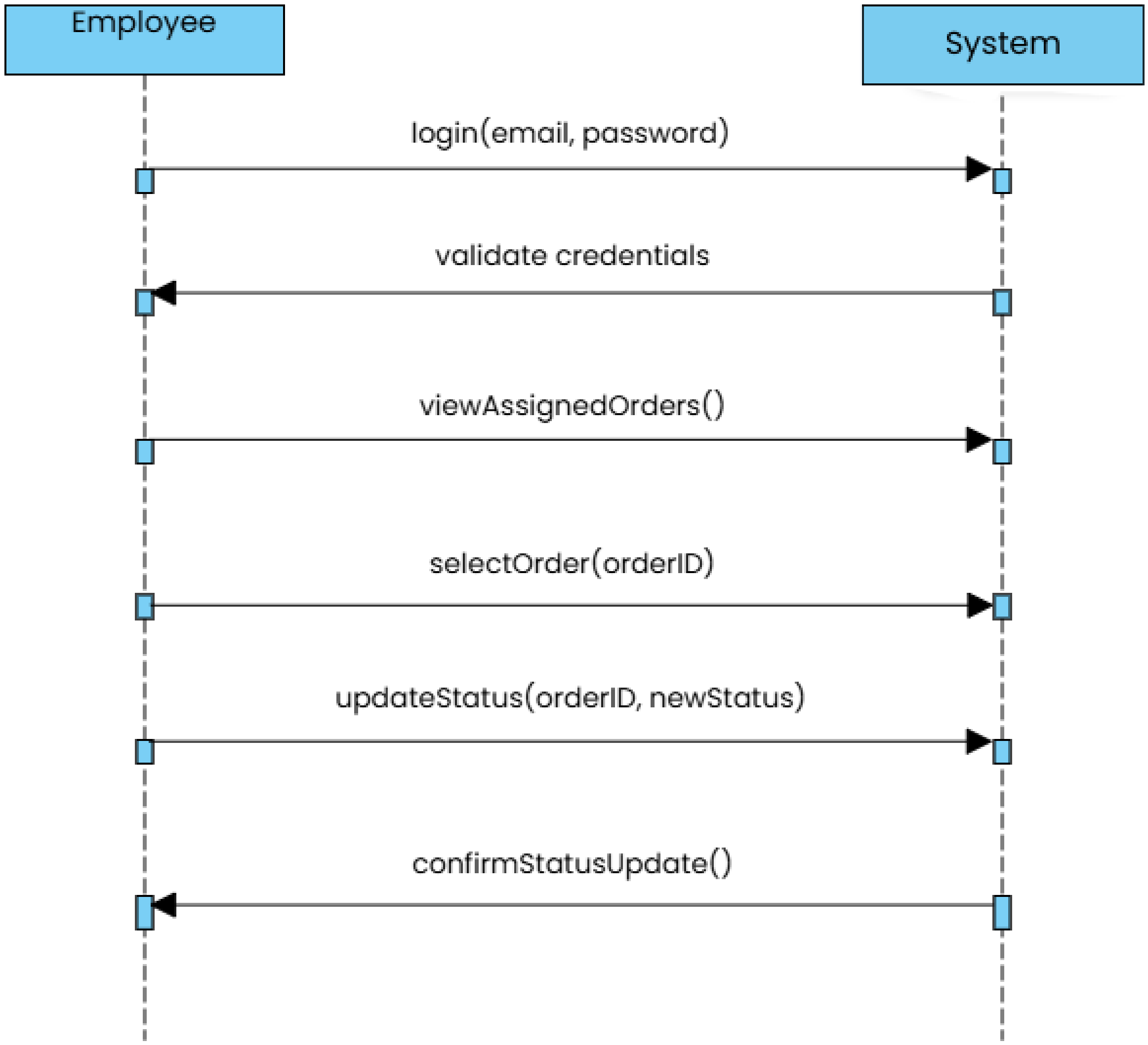

Figure 3.7.SSD for tracking order

3- Add Feedback

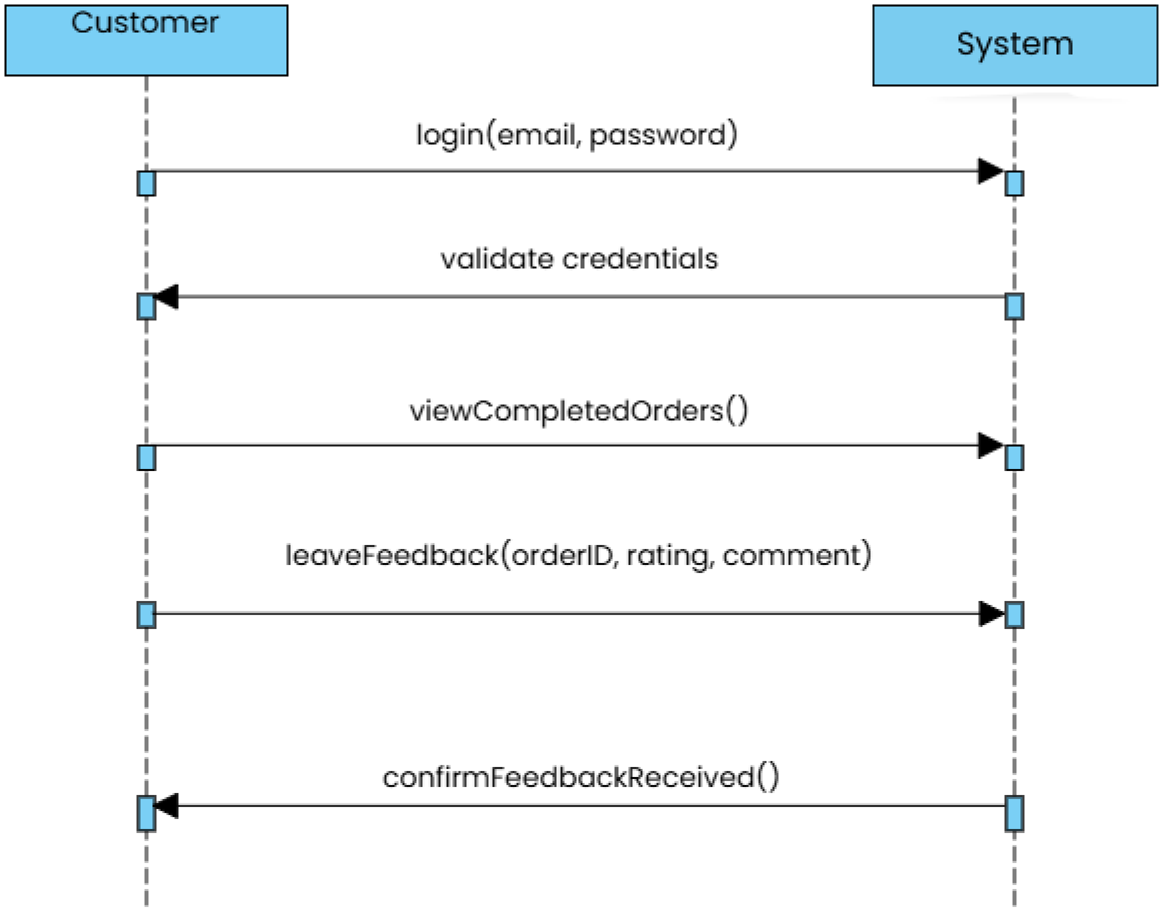

Figure 3.8.SSD for adding feedback

3.8. Sequence Diagram

Sequence diagrams, when used in conjunction with class diagrams; provide an extremely effective communication mechanism. UML sequence diagrams as shown in Fig. 3.9 are used to show how objects interact in a given situation.

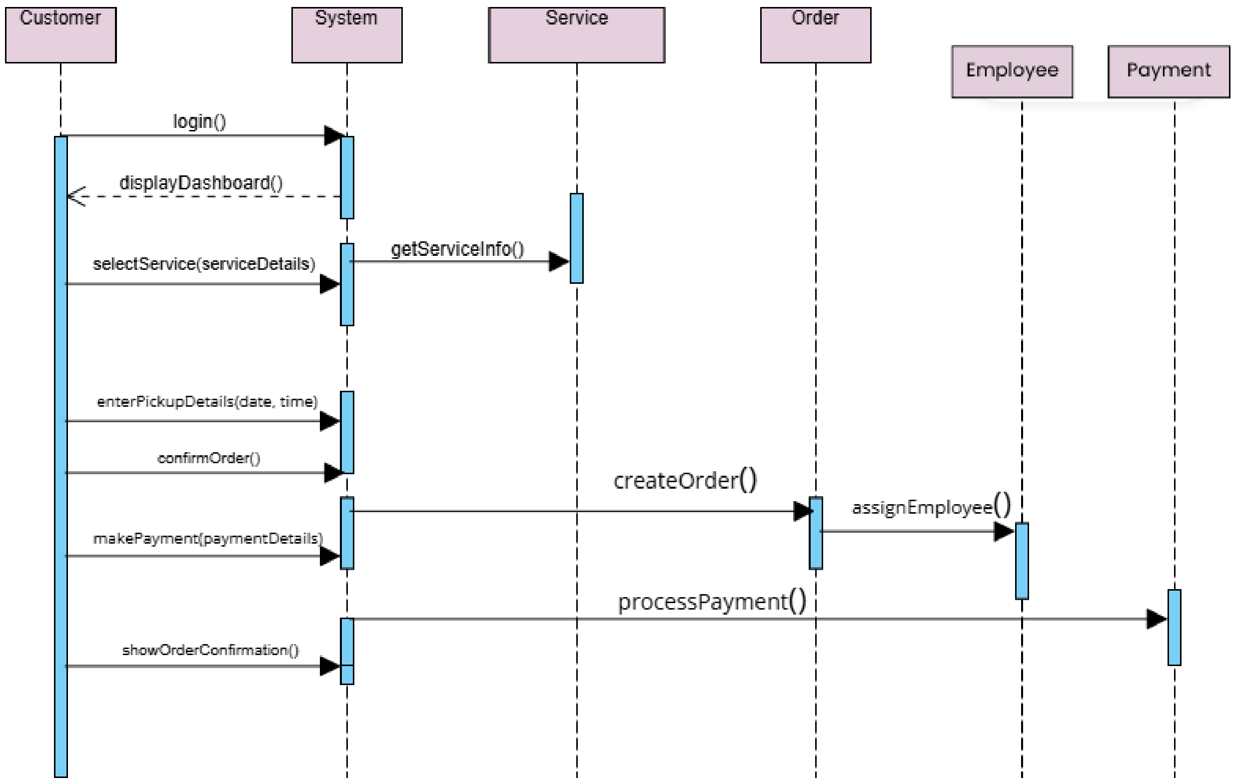

Figure 3.9. Sequence Diagram

3.9. RDM

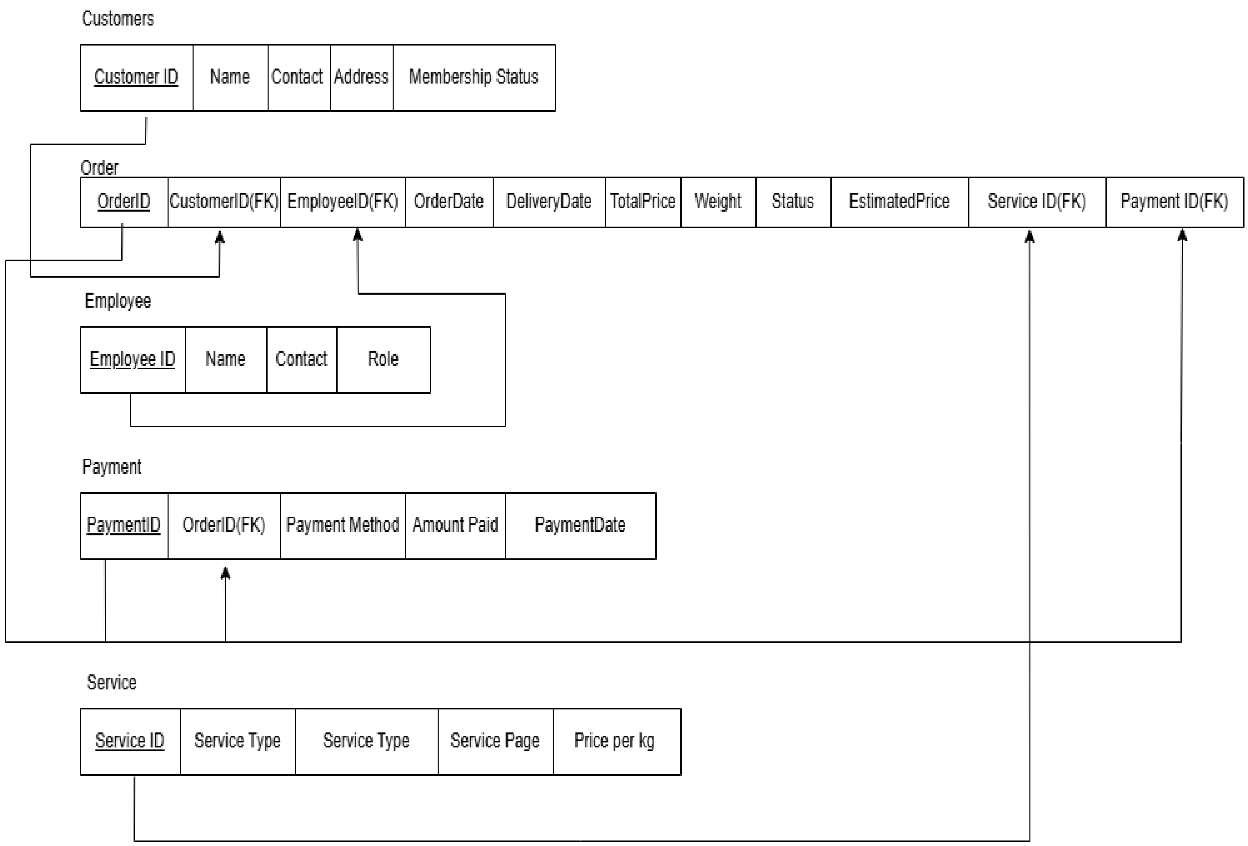

Figure 3.10.Sequence Diagram

Chapter 4

Software Development

The Implementation section is similar to the Specification and Design section in that it describes the system, but it does so at a finer level of detail, down to the code level. This section is about the realization of the concepts and ideas developed earlier. It can also describe any problems that may have arisen during implementation and how you dealt with them.

4.1. Coding Standards

Aspect	Standard/Convention Followed
1. Indentation	- Used 2-space indentation for both HTML and CSS. - All nested elements were properly indented to reflect structural hierarchy and improve readability. - Auto-formatting tools (like VS Code's "Format Document") were applied consistently.
2. Declaration	- In HTML, all elements were properly opened and closed. - In CSS, every property was followed by a semicolon and declared clearly inside curly braces. - No inline styles were used — styles were separated into CSS files.
3. Naming Convention	- Followed kebab-case for all class and ID names (e.g., `order-form`, `laundry-status`). - Names were semantic and descriptive, indicating the function or section of the webpage. - Avoided generic names like `div1`, `style1`.
4. Statement Standards	- One property per line in CSS for clarity. - Each CSS block (selector and properties) separated by a newline. - Used consistent formatting for HTML tags (e.g., `<input type="text" class="input-box">`). - Included comments for complex sections.
5. File Organization	- HTML and CSS were separated into different files (`index.html` and `style.css`). - Folder structure was neat with assets (like images) in separate directories (e.g., `images/`, `css/`). - Linked CSS using `<link rel="stylesheet" href="css/style.css">`.

4.2. Development Environment

The Laundry Management System project was developed using a simple, effective, and accessible frontend development environment. Since this is a frontend-only project, the focus was on user interface design and layout using standard web technologies.

Tools & Technologies Used:

Tool / Technology	Purpose
HTML5	Used to create the structure and layout of the web pages.
CSS3	Used for styling, layout design, and responsiveness.
Visual Studio Code (VS Code)	Chosen as the main code editor due to its lightweight performance, extensions support, and user-friendly interface.
Live Server Extension	Used within VS Code to instantly preview HTML and CSS changes in the browser.
Google Chrome (Browser)	Used for testing, debugging, and verifying responsive design with DevTools.

Deployment Setup:

- The project was developed and tested locally on a Windows machine using VS Code.
- Live Server was used to deploy a temporary localhost preview environment during development.
- Files were structured into folders (css/, images/, etc.) to simulate real-world deployment organization.
- Final deployment is done on GitHub.

📦 Packages or Third-party Libraries:

- No complex packages or dependencies were used as it was a pure HTML and CSS project.
- Fonts and icons were imported from Google Fonts and Font Awesome to enhance the UI.
- No backend frameworks or JavaScript libraries were used.

4.3. Software Description

1. Booking Form (from Booking.html)

Code Snippet:

```html
<form action="submit.html" method="post">
  <input type="text" placeholder="Your Name" required>
  <input type="text" placeholder="Address" required>
  <input type="number" placeholder="Number of Clothes" required>
  <input type="submit" value="Book Now">
</form>
```

Description:

This form allows users to book a laundry service by entering basic details like name, address, and number of clothes. The form uses the POST method and redirects to a "submit" page after completion.

2. Navigation Bar (from index.html)

Code Snippet:

```html
<nav>
  <ul>
    <li><a href="index.html">Home</a></li>
    <li><a href="services.html">Services</a></li>
    <li><a href="Pricing.html">Pricing</a></li>
    <li><a href="Booking.html">Book Now</a></li>
    <li><a href="Contact.html">Contact</a></li>
  </ul>
</nav>
```

Description:

This is the main navigation bar used across the site. It provides easy access to major pages like services, booking, and contact.

3. Basic CSS for Buttons (from Booking.css)

Code Snippet:

```css
css
input[type="submit"] {
  background-color: #4CAF50;
  color: white;
  padding: 10px;
  border: none;
  border-radius: 5px;
  cursor: pointer;
}
```

Description:

This CSS styles the "Book Now" button. It gives it a green background, white text, rounded corners, and a pointer cursor on hover.

4. Responsive Container (from contact.css)

Code Snippet:

```css
css
.container {
  width: 90%;
  margin: auto;
  max-width: 1200px;
}
```

Description:

This CSS snippet ensures the content is responsive. It centers the layout and limits the maximum width for readability across all screen sizes.